Limits to arbitrage in Indian equities

BS project presentation (EC 495)

November 2026

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Guide: Prof. Aditi Chaubal, Department of Economics, IIT Bombay

Agenda

Executive summary

The question and our approach

- Why India, how the F&O ban works, the model and the frozen test plan

The F&O ban

- Trading, correction and the level of the basis when a ban starts

The band: costs, capital and position limits

- How wide the band is and what moves it

Short-sale constraints

- F&O entry and a constraint index for every listed stock

Conclusion and policy

Agenda

Executive summary

Executive summary

Limits to arbitrage bind hard on trading in Indian equities. On prices they bind less, and differently, than the textbook channel predicts.

Bans bind on trading. In a ban, futures volume falls by about two log points. Once violations were judged on intraday positions (from March 2020), every futures trade in a ban closed positions.

Correction is not one-sided as predicted. Rich futures revert faster in a ban, mostly in its first days, and the basis falls about 6 bp when a ban starts before recovering within a week. Neither appears at false ban dates.

Costs shape a wide band; capital leaves weak traces. The basis moves in a band of about -35 to +24 bp; a new cash-and-carry round trip (27 to 31 bp) lies in its outer tail. Inflows to arbitrage funds go with a lower upper tail, weakly.

Short-sale access speeds price adjustment. After a stock enters the F&O segment its price delay falls by 0.11. A constraint index built from frictions predicts slower adjustment across all stocks, also after a size control.

The 2025 reform met its aim. Against the old rule applied to the same stocks, the October 2025 rules cut ban days by about three quarters.

Source: NSE archives, 2010 to 2026; author's analysis

Three of four primary tests hold as specified

The four pre-specified primary hypotheses, corrected jointly (Holm, 5%)

Hypothesis	Prediction	Result	Holm p	Verdict
H8 Trading in a ban	Futures volume falls; trades only close positions	-2.00 log points (s.e. 0.11)	7e-41	**Supported**
H3 Costs and the band	Upper edge rises with each tax increase	+6.96 bp (s.e. 0.61); weak against placebos	6e-24	**Supported**
H6 Short-sale access	F&O entry lowers delay and bad-news drift	Delay -0.109 (s.e. 0.023); drift the other way	1e-5	**Supported (delay)**
H7 One-sided correction	Rich futures revert more slowly in a ban	-0.089 (s.e. 0.037): faster, not slower	1	**Rejected**

Ten secondary tests were reported without adjustment and read as a pattern; three are below 0.05.

Every detail the plan left open was fixed in a dated log before the test that needed it.

The failed prediction is reported in full: rich futures revert faster in a ban

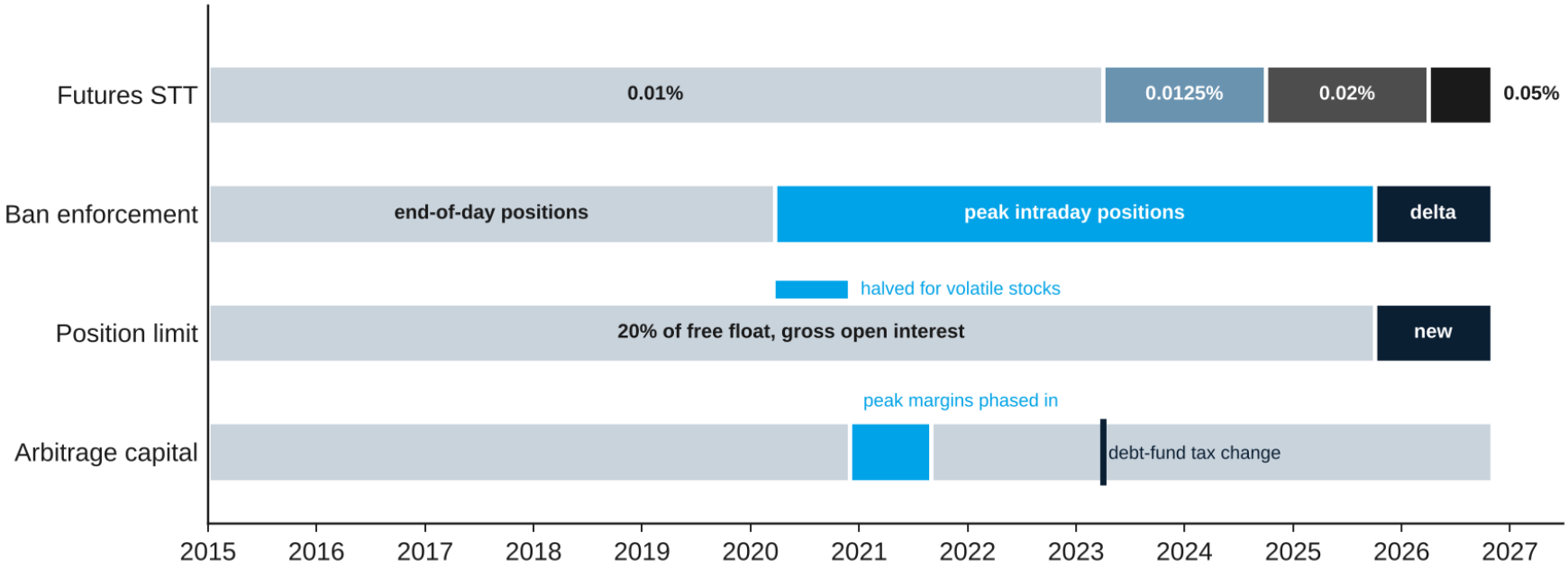
Source: NSE archives; author's analysis

Agenda

The question and our approach

India changes almost every limit on a known date

Futures STT, ban enforcement, the position-limit rule and arbitrage capital, 2015 to 2026

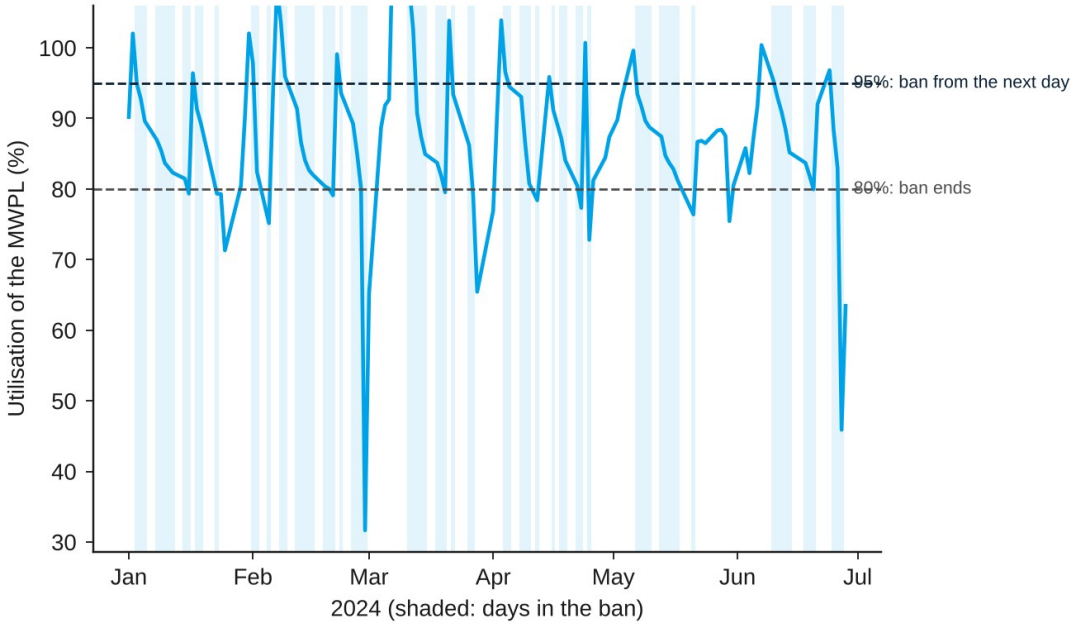


Dated rule changes let us measure each limit instead of assuming it

Source: SEBI circulars and press releases; NSE circulars; AMFI

The F&O ban lets every trader only reduce positions

Utilisation of the position limit for SAIL, the most-banned stock, January to June 2024



How the rule works

Enter the day after open interest exceeds 95% of the market-wide position limit

Leave after it falls to 80%

In between, everyone may only reduce positions, in all expiries

1,788 spells in 141 stocks, 2015 to October 2026

To my knowledge, no study had estimated what this rule does to prices

Source: NSE archives; author's analysis

We fixed the tests before looking at outcomes

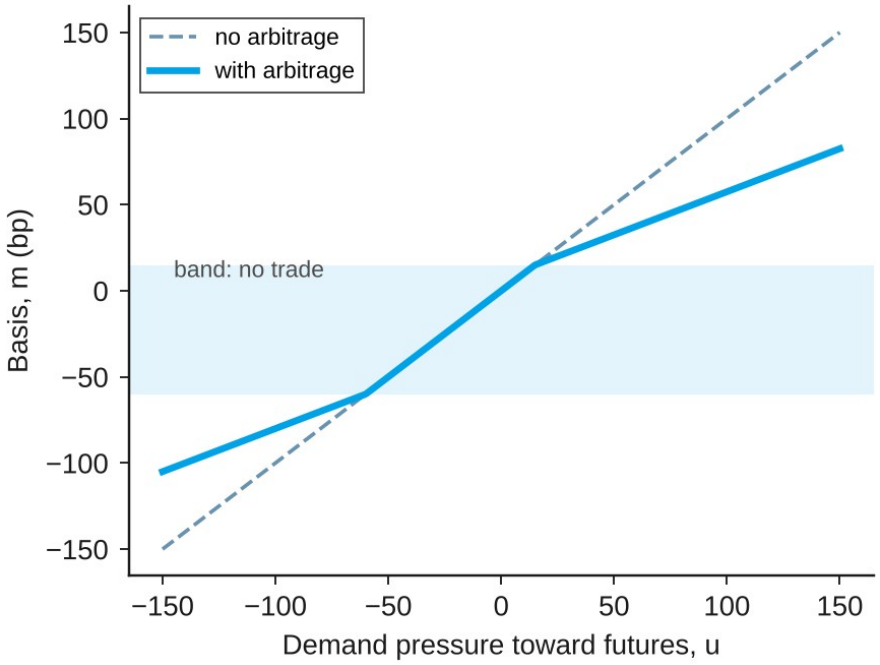
The five safeguards of the project

		What we did	Why
1	**Model**	One model; **every limit a parameter**; each prediction with what would contradict it	Hypotheses are derived, not fitted
2	**Frozen plan**	Tests and signs **fixed with a git time stamp** on 5 November 2026	No choosing specifications after seeing results
3	**Blind data**	Built from **public NSE files**; treatment and outcome code kept apart	No peeking while the data are built
4	**Estimator tests**	Each estimator **recovers known effects** on simulated data	Methods checked before real data
5	**Full reporting**	Primary family **Holm-corrected**; nulls and contradictions kept	The failed prediction stays in

Source: NSE archives; author's analysis

Costs set the band; a ban blocks two things

Equilibrium basis against the demand pressure of outside traders, with illustrative costs



Band	Arbitrage starts only outside **[-c-, c+]** • Costs set the edges; capital and risk set how far the basis travels beyond them
Ban: arbitrage	• Arbitrageurs may only reduce positions • **Rich futures are no longer corrected**
Ban: demand	• New futures demand is blocked; part moves to the stock • **This lowers the basis**

So the ban's effect on the level has no fixed sign: it depends on what is blocked

Source: author's model (thesis, Chapter 4 and Appendix C); illustrative parameters, not estimates

Each hypothesis states what would contradict it

Predictions derived from the model; the first four form the primary family

	Prediction	Contradicted if
H8	In a ban futures volume falls and every trade closes positions	Futures activity unchanged
H7	In a ban rich futures revert more slowly, cheap ones as fast or faster	No asymmetry, or the opposite
H3	Band edges move with the STT, more for larger increases	Edges unrelated to cost changes
H6	F&O entry lowers price delay and bad-news drift; exit raises them	No change or the opposite
Level	The basis level at ban start follows the composition of open interest	Sign unrelated to composition
H9	The upper tail grows smoothly as utilisation nears 95%	Jump only at 95%
H1, H4, H5	Wider bands with frictions; scarcer capital widens; overpricing anomalies stronger where shorting is hard	No relation; symmetric effects

Source: frozen pre-analysis plan (git tag pap-v1, 5 November 2026)

The data come entirely from public NSE files

Sources, coverage and the main checks

Bans and limits	• Daily open interest, position limits and ban lists, February 2010 to October 2026 • Next-day flags match the official ban list for 99.9% of stock-days
Prices and basis	• Futures and cash bhavcopies 2015 to 2026; basis from volume-weighted prices • Settlement prices avoided: they are set at cost of carry on thin days
Events	• Results dated from the exchange's first dissemination time • 239 entries into and 171 exits from the F&O segment
Capital, lending	• AMFI arbitrage-fund flows and assets, April 2018 to September 2026 • Daily securities-lending trades, 2015 to 2026

1,788
ban spells

486,040
stock-days of basis

80,042
stock-quarters in the index

3,310
stocks in the cash market

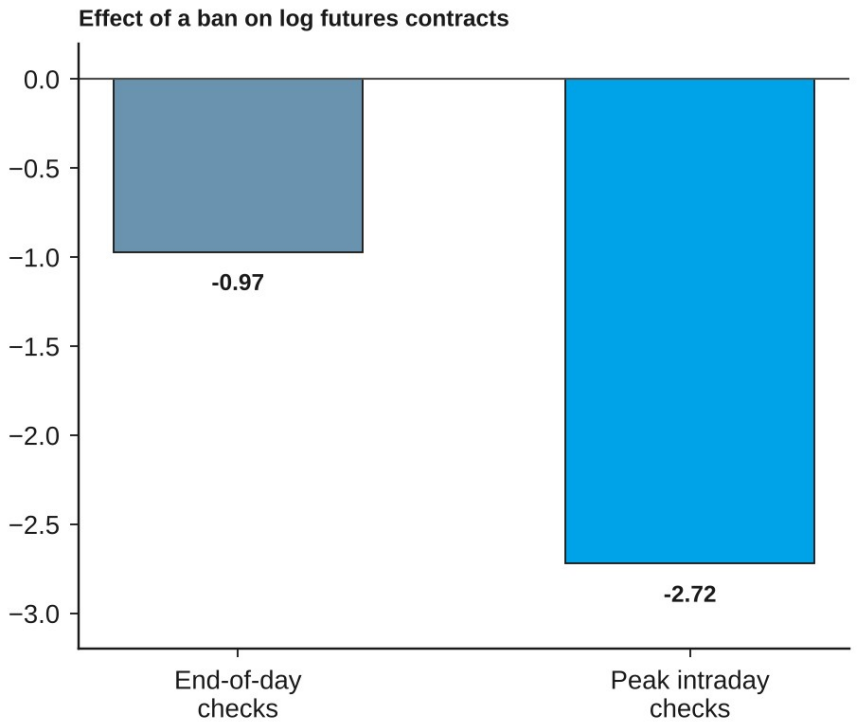
Source: NSE archives; AMFI; OECD via FRED; author's analysis

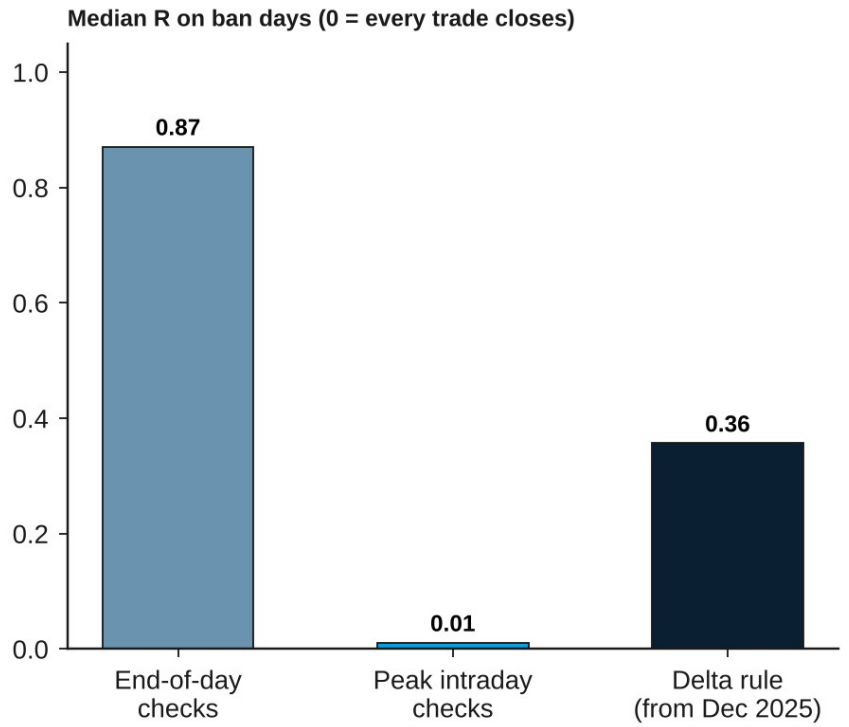
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The F&O ban

In a ban, futures trading collapses

Effect of a ban on log futures contracts, and the median R on ban days, by enforcement regime



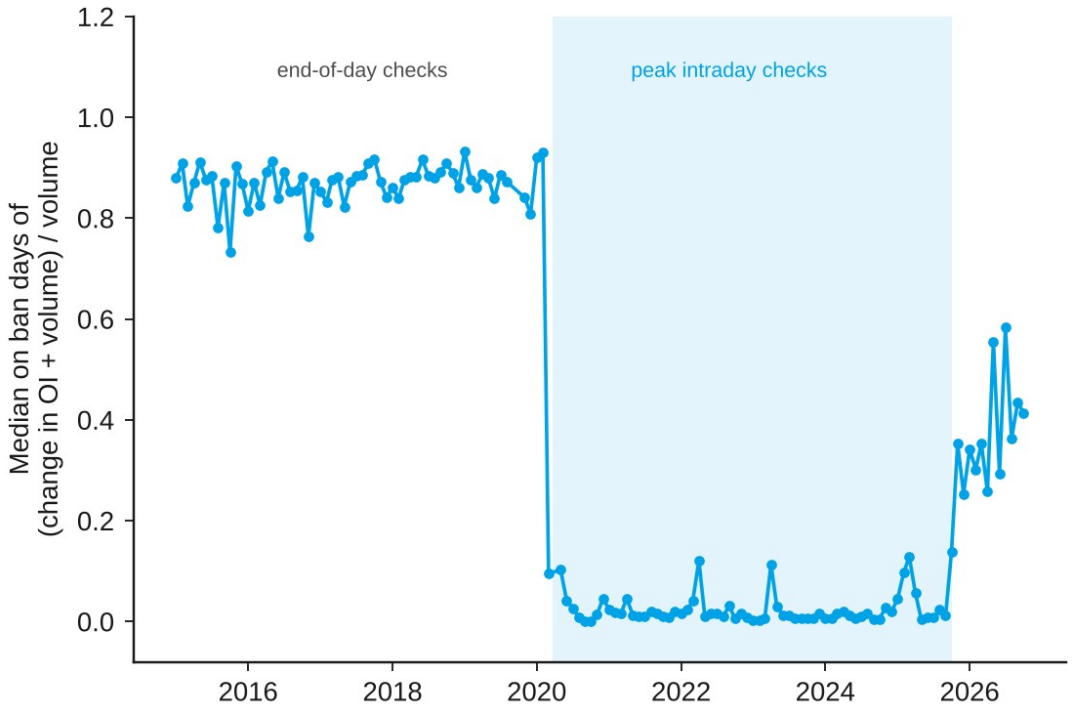


Futures volume falls by about 87% in a ban; cash trading rises and turns toward delivery

Source: NSE archives; author's analysis

Intraday checks made the rule bind trade by trade

Monthly median on ban days of R = (change in open interest + volume) / volume



What the chart shows

To 22 March 2020 violations were judged on end-of-day positions: median R 0.87

From 23 March 2020 on peak intraday positions: median R 0.011

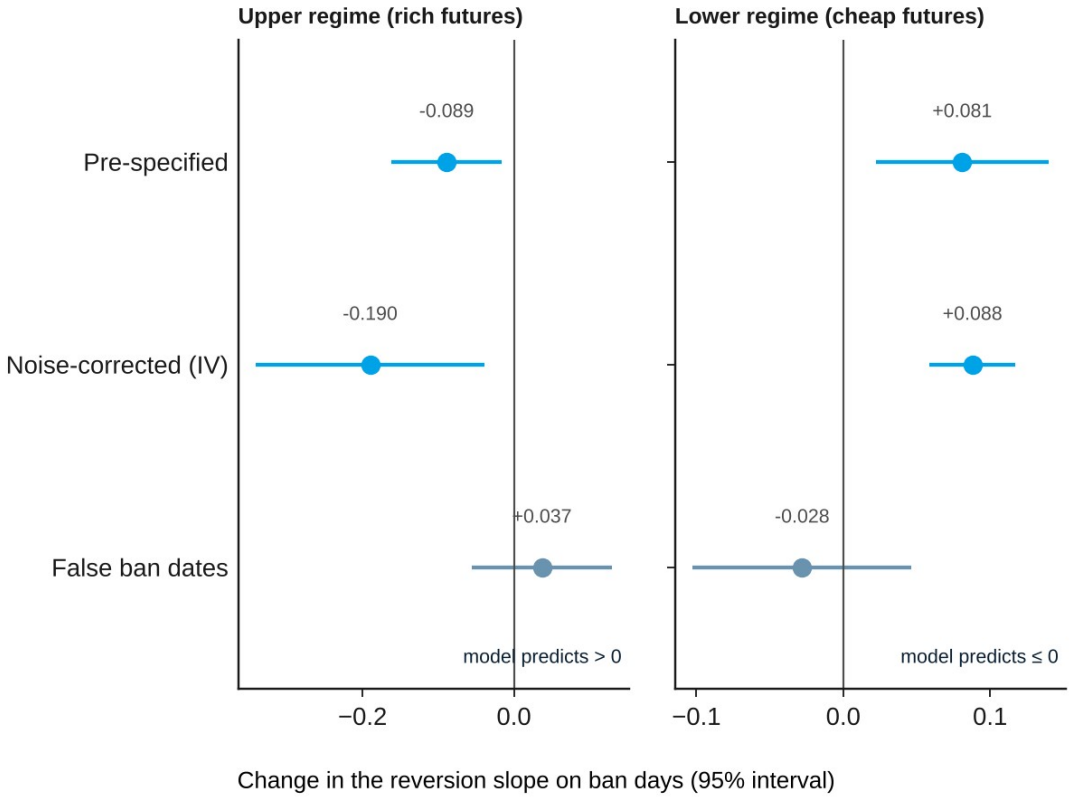
From October 2025 the delta rule allows offsetting trades: 0.19 to 0.36

The rule changes trading as designed only when positions are checked during the day

Source: NSE archives; author's analysis

Correction is not one-sided: rich futures revert faster

Change in the reversion slope on ban days, by regime, with 95% intervals



Why it matters

Predicted: slower reversion of rich futures (above 0)

Found: -0.089 (s.e. 0.037); cheap futures revert more slowly

Robust: -0.190 when instrumented; +0.037 at false ban dates

Timing: concentrated in the first five days of a spell

The pre-specified prediction failed, with the opposite sign

Source: NSE archives; author's analysis

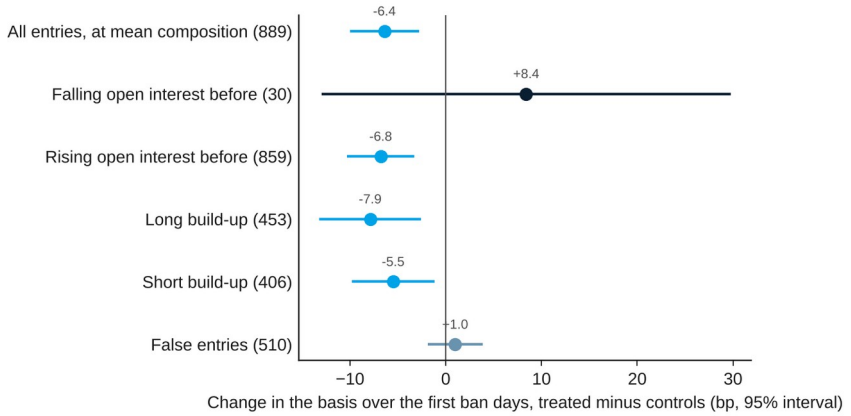
The basis falls when a ban starts, then recovers

Change in the basis around ban entry, treated minus matched controls (bp)

About -6 bp in the first days, more after fast build-ups



Both long and short build-ups; nothing at false entries



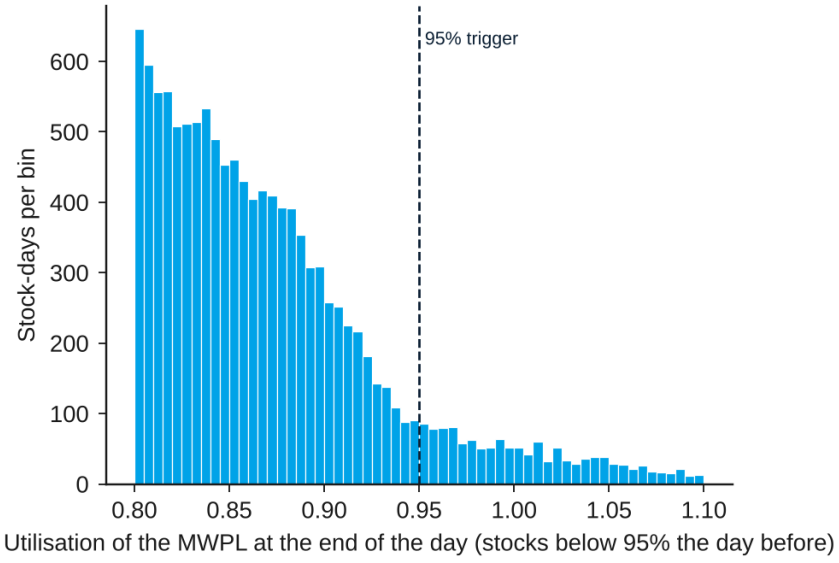
Blocked new futures demand fits both facts, a reading formed after the results

Source: NSE archives; author's analysis

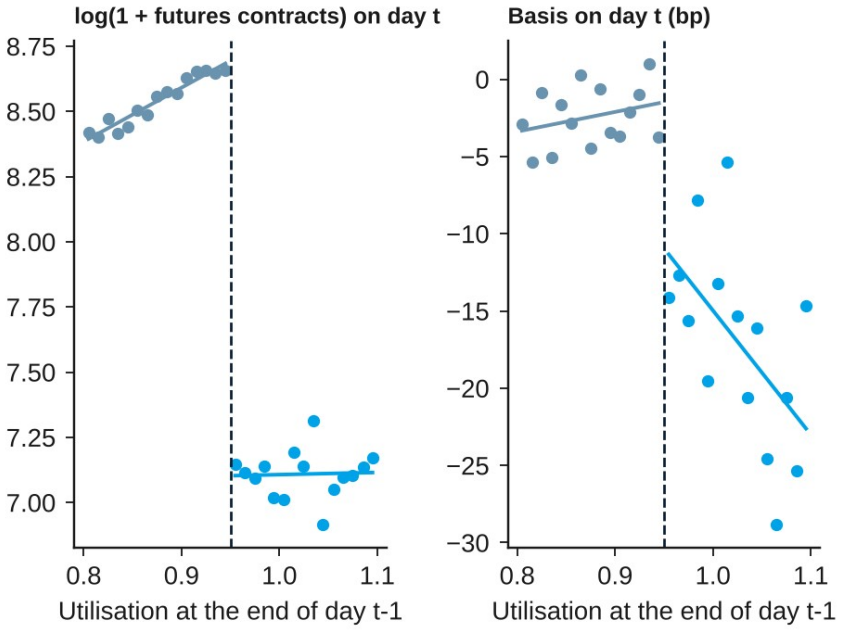
Traders do not hold back below the 95% trigger

Utilisation around the trigger, and outcomes on either side of it

No bunching below 95% (density test p = 0.86)



Futures trading drops at 95%; the basis is lower above it



The trigger works mechanically; the basis gap is already there the day it is crossed

Source: NSE archives; author's analysis

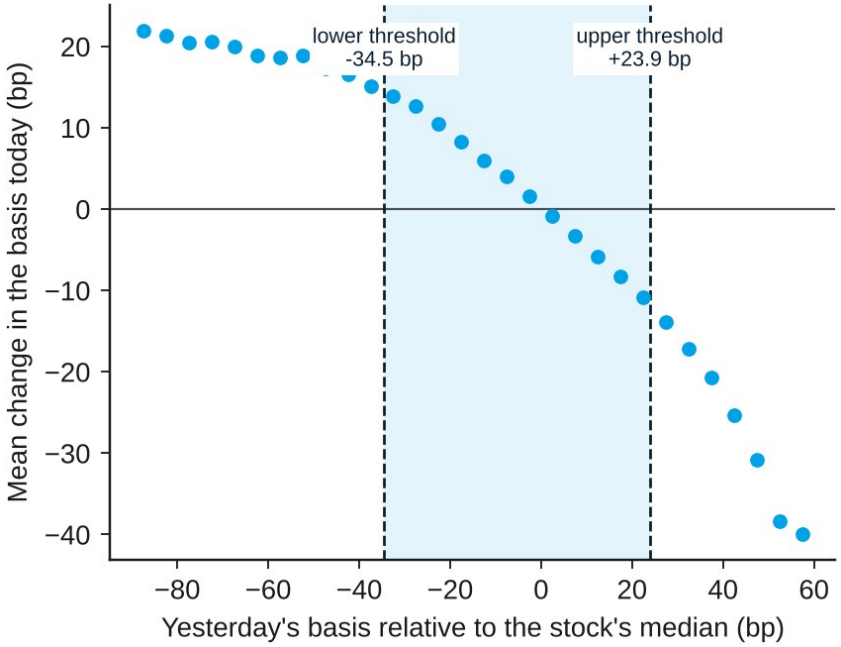
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The band: costs, capital and position limits

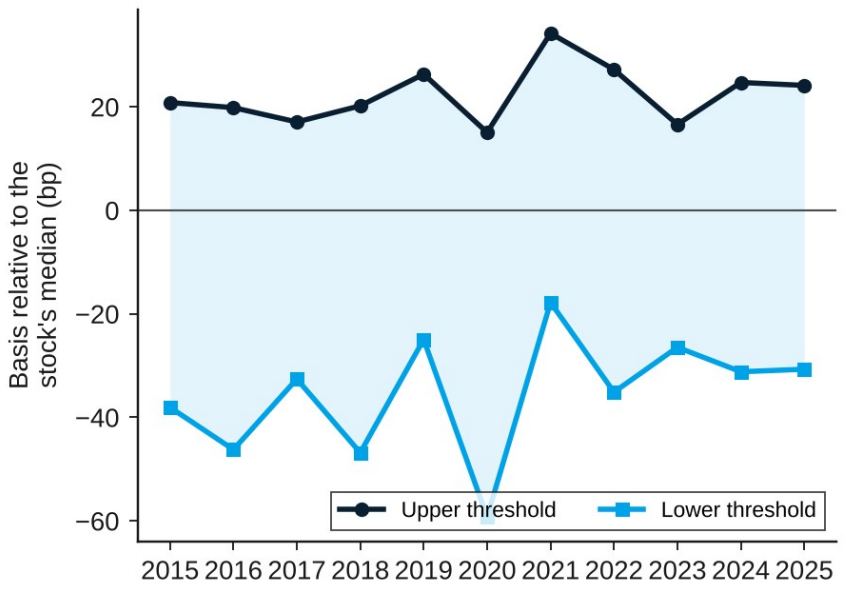
The basis moves in a wide, asymmetric band

Pooled threshold model, non-ban days, 2015 to 2026 (473,110 observations, 360 stocks)

Reversion grows with the deviation; band -34.5 to +23.9 bp



Stable across years, apart from March 2020



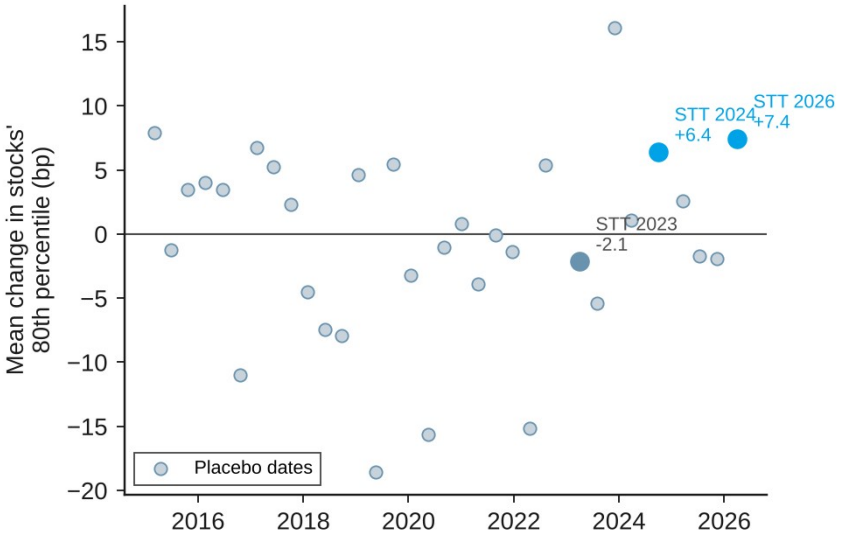
Much daily reversion is price noise; where the band lies is the robust part

Source: NSE archives; author's analysis

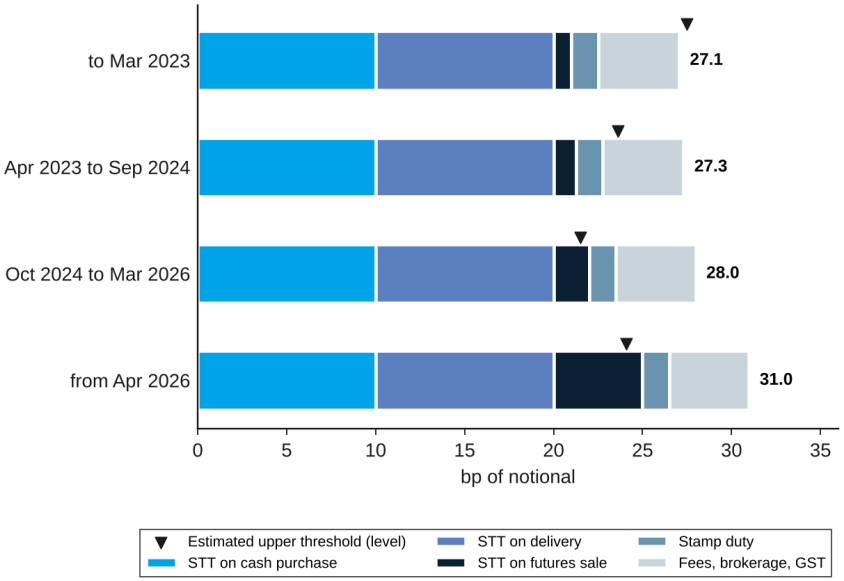
Tax rises lifted the whole distribution

Change in stocks' 80th percentile at STT dates, and the bottom-up cost of a new position

Large changes, but within the range of placebo dates



A new round trip costs 27 to 31 bp, above the upper edge



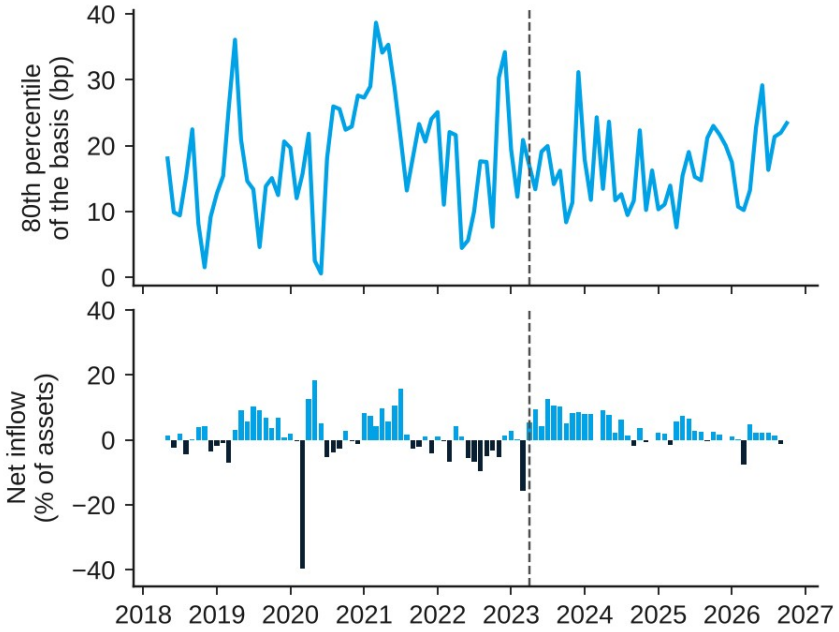
New capital enters only at large gaps; funds that roll lean against small ones (4 to 8 bp)

Source: NSE archives; author's analysis

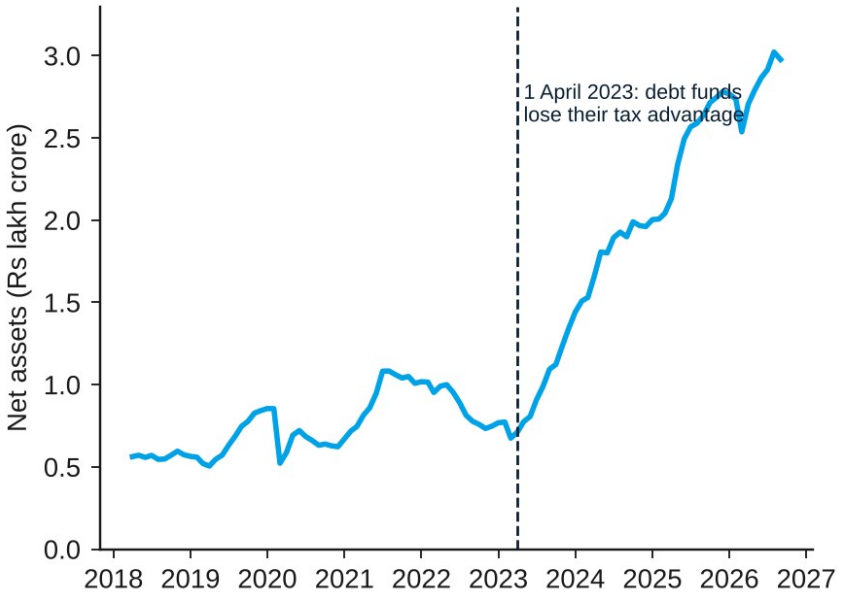
Arbitrage capital leaves weak traces

Arbitrage-fund flows, assets and the upper tail of the basis, monthly

Inflows go with a lower upper tail next month (p = 0.11)



Fund assets quadrupled after debt funds lost their tax edge



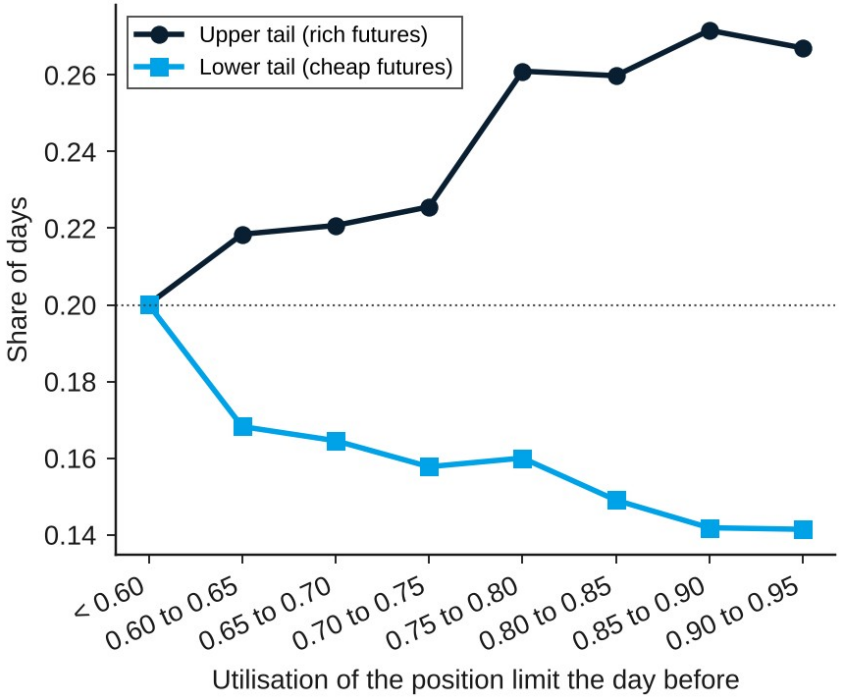
The link is clearer month on month and before 2023, and invisible after

Source: AMFI monthly reports; NSE archives; author's analysis

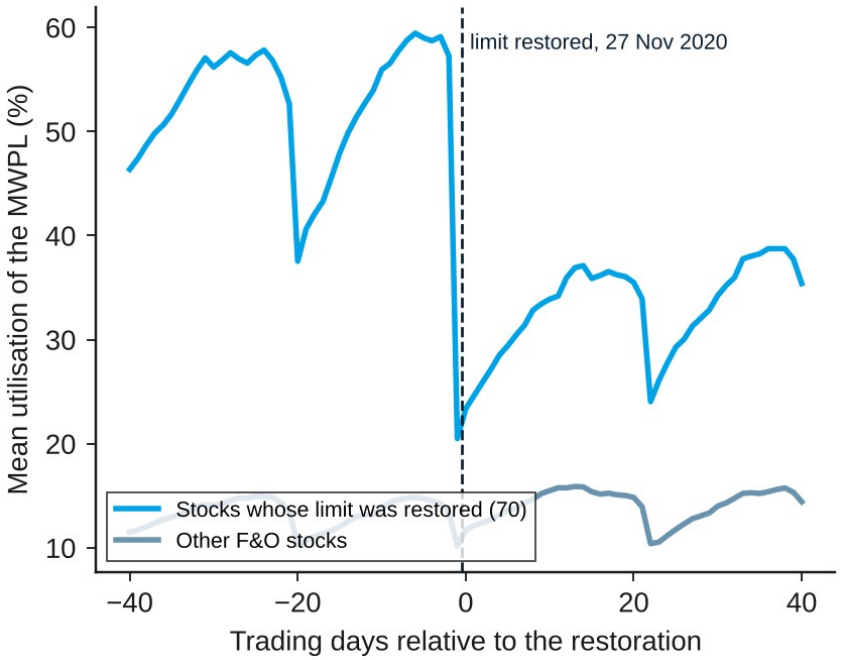
Rich futures become more common near the trigger

Upper-tail share by utilisation, and the 2020 restoration of halved limits

The upper-tail share rises from 0.20 to 0.27 (p = 0.019)



Restoring halved limits cut utilisation at once

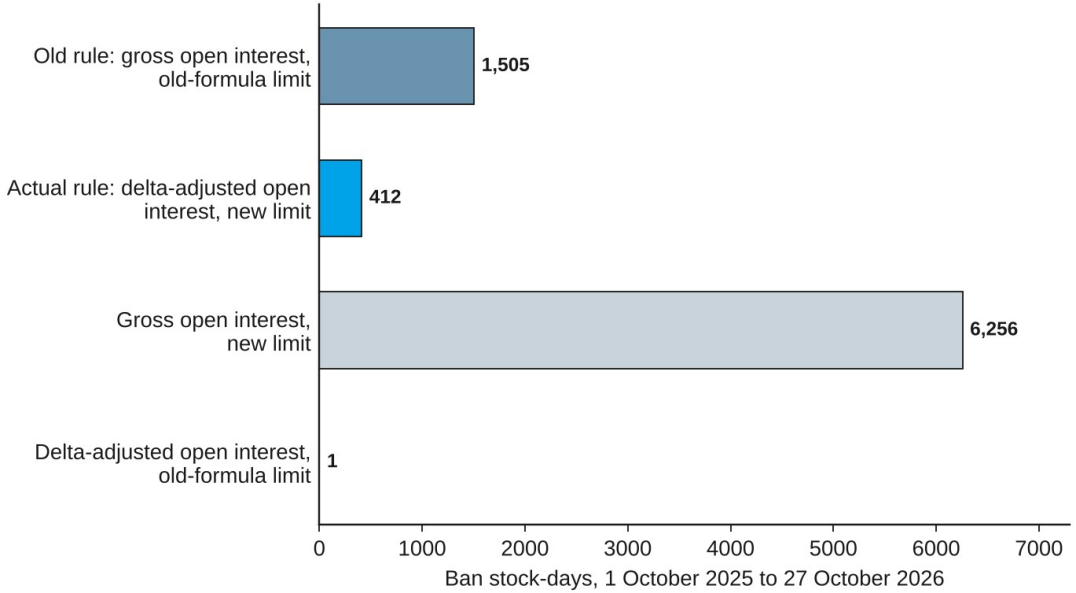


The level of the limit matters for prices even before a ban starts

Source: NSE archives; author's analysis

The 2025 rules cut ban days by about three quarters

Ban stock-days, 1 October 2025 to 27 October 2026, actual and counterfactual rules



What the reform did

Aim: SEBI set out to reduce 'spurious' bans

Before 2025 bans came from rising open interest; only 0.4% from a falling limit

Delta-adjusted open interest removes almost all bans; the lower limit alone would have quadrupled them

Net: 412 ban days against 1,505 under the old rule

The reform met its stated aim

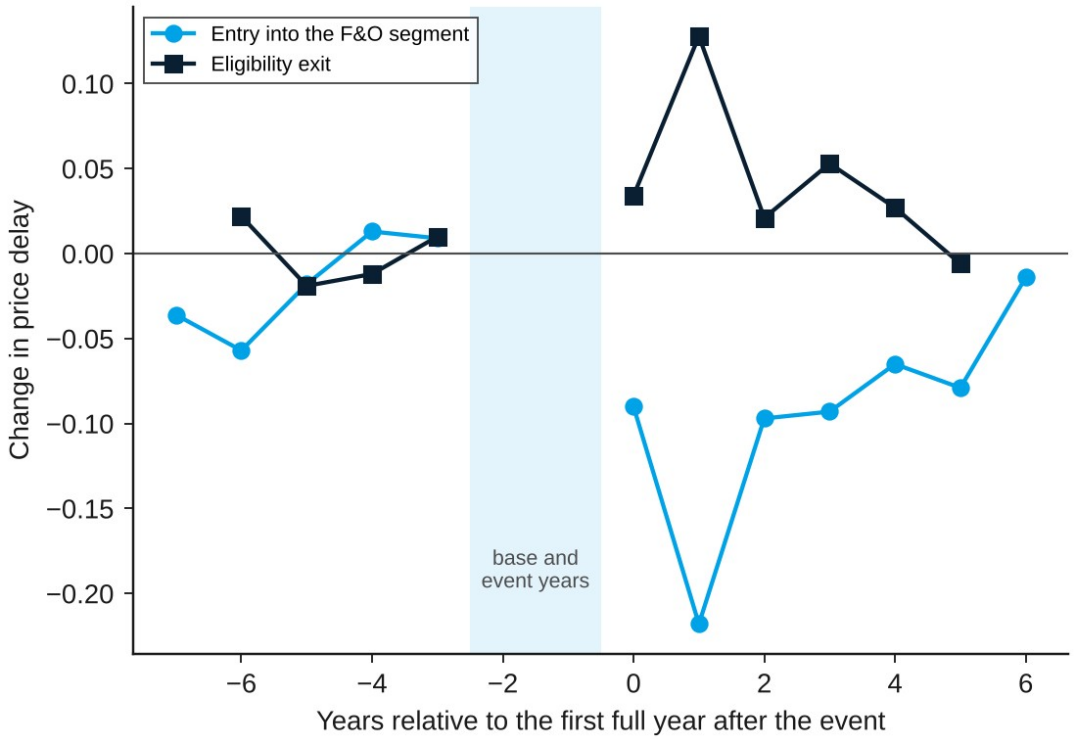
Source: NSE archives; author's analysis

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Short-sale constraints

F&O entry speeds up how prices respond to news

Change in price delay by year relative to the first full year after entry or exit



Component	Estimate	p
Entry: price delay	-0.109	1e-6
Entry: bad-news drift	-0.040	0.99
Exit: price delay	+0.053	0.09
Exit: bad-news drift	+0.014	0.77

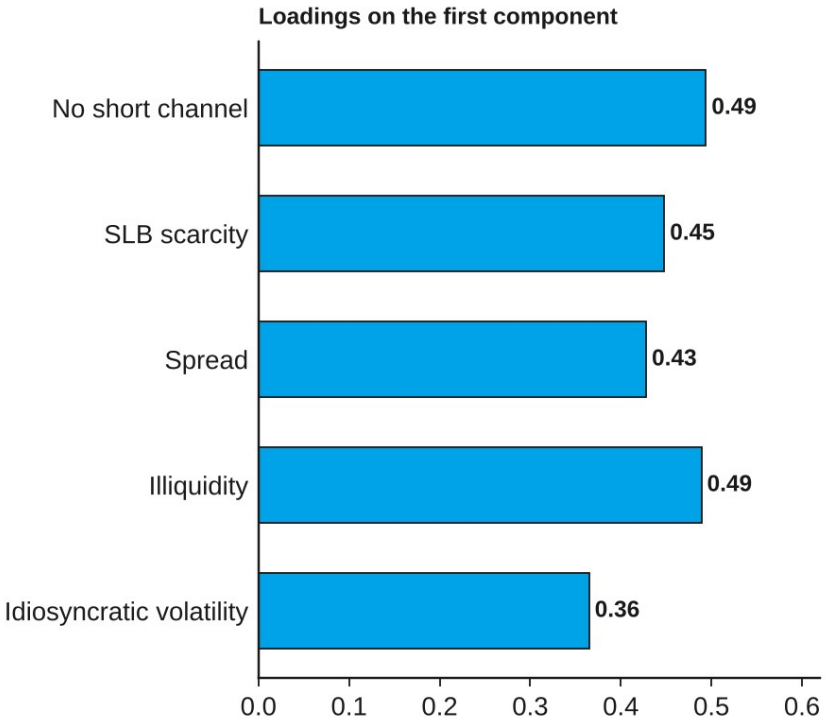
Prediction: entry lowers delay and drift; exit raises them

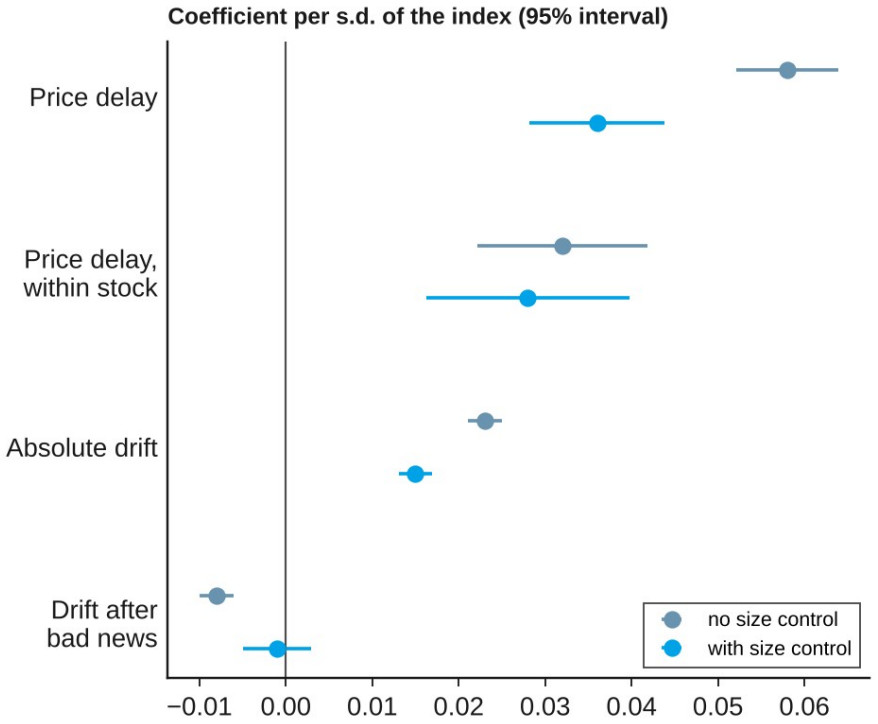
Delay falls by 0.11 after entry (median 0.24 before); drift after bad news does not weaken

Source: NSE archives; author's analysis

Constrained stocks adjust more slowly, market-wide

A quarterly index from five frictions for all NSE stocks: loadings and validation



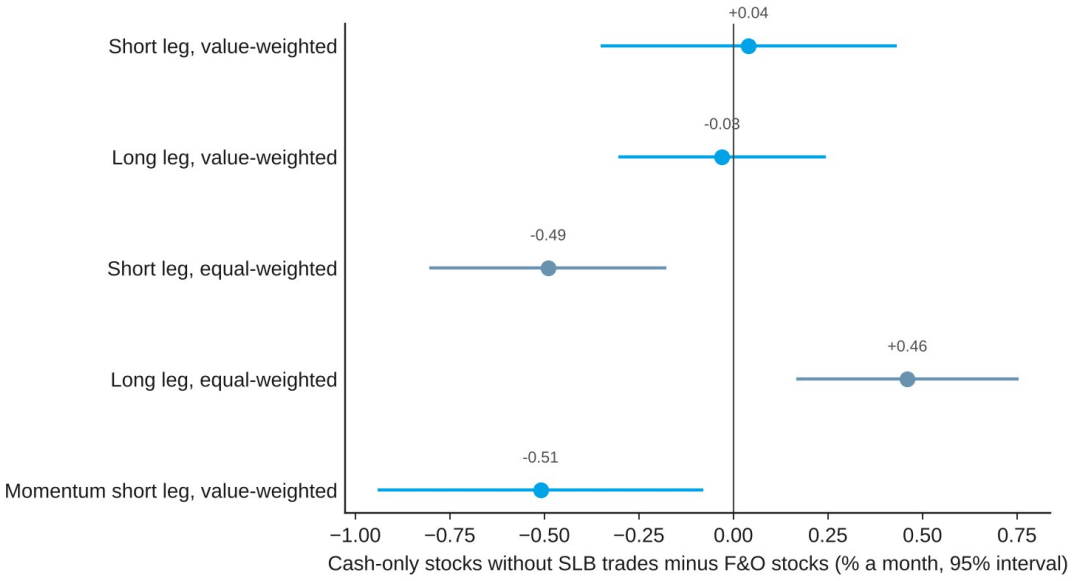


The link with price delay survives a size control; the bad-news link is a size effect

Source: NSE bhavcopies and SLB files; AMFI market capitalisation; author's analysis

Anomalies are not one-sided where shorting is hard

Cash-only stocks without lending minus F&O stocks, % a month



What the chart shows

Value-weighted, the overpriced side does no worse where shorting is hardest

Equal-weighted, anomalies are stronger among small cash-only stocks, on both sides

Momentum alone fits the prediction (-0.51%)

Groups differ greatly in size, which limits what this comparison can show

Source: NSE archives; author's analysis

Agenda

Conclusion and policy

Limits bind hard on trading, less cleanly on prices

What each limit does, and how strong the evidence is

Limit	On trading	On prices	Main evidence
Position limits	**Strong**	Real, not one-sided	Volume -2 log points; basis -6 bp at ban start; 2025 rules cut bans by three quarters
Short-sale constraints	Not tested	**Speed of adjustment**	Delay -0.11 after F&O entry; no effect on bad-news drift or overpricing
Trading costs	Not tested	Set the band	Band -35 to +24 bp; new round trip in the outer tail; edges not one-for-one with the tax
Arbitrage capital	Not tested	Weak	Inflows go with a lower upper tail (p = 0.11); clearer in differences

Position limits matter most; costs shape the band; capital is a weak force

Source: NSE archives; author's analysis

Bans restrain demand at the cost of arbitrage

What the evidence shows, and what it implies for the rules

What the evidence shows

- A ban cuts futures volume by almost 90%
- Its price effect appears at the start of a ban and fades within a week
- The rule bites as designed only under intraday checks
- The 2025 rules cut ban days by about three quarters

What it implies

- A ban blocks the speculative demand it targets, and the arbitrage that keeps prices together
- Enforcement matters as much as the threshold
- Measuring open interest in delta terms removes avoidable bans
- F&O access has a measurable benefit: faster price adjustment

-87%
futures volume in a ban

~1 week
for the price effect to fade

-73%
ban days, 2025 rules against old

93%
of individual F&O traders lost money, FY22 to FY24

Limits restrain a large speculative demand, and arbitrage pays part of the price

Source: author's analysis; SEBI press release 22/2024; SEBI circulars 2024/132 and 2025/79

Four limitations we know about

Limitation	What could go wrong	...and what I did
Daily data	Intraday arbitrage is out of scope, and noise inflates measured reversion	Noise-corrected estimates reported alongside
Inventory unobserved	The model's arbitrage book is not seen in public data	Composition of open interest used as a proxy
Short post-reform sample	Only 44 ban spells since October 2025	Post-reform results labelled low-powered
Size confounds	Constrained stocks are much smaller	Size controls; F&O entry as a within-stock design

Intraday data and a longer post-reform sample would sharpen the ban tests

Source: NSE archives; author's analysis

Limits to arbitrage in Indian equities

Thank you

November 2026

Prince Kumar (23B3032)

Guide: Prof. Aditi Chaubal, Department of Economics, IIT Bombay

Agenda

Appendix

All pre-specified tests

Primary family (Holm) and secondary tests (unadjusted)

Test	Result	Comment
H8 trading	Supported (p < 1e-10)	Identity holds under intraday enforcement
H3 costs	Supported (p < 1e-10)	Weak against placebos; mostly a common shift
H6 short-sale access	Supported (p = 5e-6)	Through price delay; drift goes the other way
H7 one-sided correction	Rejected, opposite sign	Survives a noise correction
Level effect	p = 0.083	Signs as predicted; median split p = 0.025
Density, RD at 95%	p = 0.86; -12.9 bp	No bunching; gap present before the ban
Hysteresis	+0.4 bp, p = 0.64	No extra level shift
H9 anticipation, reform	p = 0.019; p = 0.11	Upper tail grows toward 95%; reform right sign
H2, H1, H4, H5	p = 0.59, 0.002, 0.11, 0.57	No power; across stocks only; weak; not supported

Source: author's analysis; thesis, Table 9.1

How the data were built and checked

Bans	• Next-day flags agree with the official list for 11,753 of 11,764 flags • 95% entry and 80% exit rules hold for 99.4% and 99.7% of spells
Basis	• Volume-weighted futures and cash prices, net of dividends and carry • Contracts with dividends or corporate events in their life excluded
Results dates	• Earliest of the exchange's announcement and the filing; board meetings fill gaps • Covers 98% to 100% of F&O stock-quarters
Reproduction	• One script sequence produces every estimate; a full rerun reproduced every number • Each estimator first recovered known effects on simulated data

Source: author's analysis; thesis, Chapter 5 and Appendix B

The model's eleven results map onto the tests

Result	Statement	Tested by
1-2 Band	Arbitrage only outside [-c-, c+]; costs set the edges, capital the passthrough	H3, H4, band chapter
3 Inventory	The no-trade region moves with the existing book	Estimator tests
4 One-sided	In a ban rich futures are not corrected; cheap ones are while inventory lasts	H7
5-6 Demand, level	Blocked new demand reverses its effect; the level effect has no fixed sign	Level effect
7 Trading	Contract-by-contract enforcement makes every trade close positions	H8
8-9 Anticipation	The upper edge rises toward 95%; the limit is not an instrument for bans	H9, density, RD
10-11 Short sales	Bad news slower without a short channel; Miller overpricing	H2, H5, H6

Source: author's model (thesis, Appendix C)